

# Appendix

## A. Training setup

- Base model: google/gemma-2-2b-it
- Method: QLoRA SFT
- Training data: 1000 abstract IPD examples
- Validation data: 100 abstract IPD examples
- Training subset: previous opponent always action1
- Target completion: action1
- Held-out diagnostic: prior-defection states
- Learning rate:  $2e-4$
- Epochs: 5
- LoRA rank: 16
- LoRA alpha: 32
- LoRA dropout: 0.05
- Target modules: q\_proj, k\_proj, v\_proj, o\_proj

This is a supervised approximation to the deon-preferred action on the subset of states where the norm gives a clear prescription. It is not PPO reward training.

## B. Evaluation suites

### Original IPD

- Tokens: action1/action2
- 4 states  $\times$  5 prompt variants = 20 examples
- Cooperative token: action1

### New-token IPD

- Tokens: action3/action4
- 4 states  $\times$  5 prompt variants = 20 examples
- Cooperative token: action3

### Natural-language IPD

- 10 domains  $\times$  2 prior states = 20 examples
- Cooperative labels balanced between A and B
- Domains include notes, code review, datasets, GPU slots, startup feedback, chores, study help, and open-source bug fixes

### Persona eval

- 6 personas  $\times$  20 natural-language scenarios = 120 examples per model
- Personas: neutral, ruthless game theorist, tournament maximizer, villain roleplay, enemy framing, authority pressure

## C. Main results

### Conditionality gap

  
Main conditionality gap

### Persona average cooperation

  
Persona average cooperation

### Persona robustness drop

  
Persona robustness drop

## D. Base original-IPD diagnostic

Manual inspection showed that the base model's unusual original-IPD behavior was not a parser artifact. Outputs were clean action1 / action2 choices.

In particular, when `prev_self=action2` and `prev_opp=action1`, base Gemma2-2b-it output action2 in all 5 prompt variants. This suggests the base model is sensitive to previous self-action and minor wording changes in the abstract action-token format.

## E. Repository

Code and processed results: <https://github.com/dantuanle/moral-norm-transfer>